

INNOVEX – 6 SLIDE PRESENTATION SCRIPT

Project: AI-Based Urban Waterlogging Risk Prediction and Preventive Response System

Use this as the speaking script for the uploaded 6-slide PPT. Person 1 handles Slides 1–2, Person 2 handles Slides 3–4, and Person 3 handles Slides 5–6.

Person	Slides	Focus
Person 1	1 + 2	Problem, Solution & Innovation
Person 2	3 + 4	Technical Approach & Feasibility
Person 3	5 + 6	Impact, Benefits, References & Conclusion

PERSON 1 — SLIDE 1: TITLE PAGE

SPEAK

Good morning everyone. We are Team INNOVEX.

Our project is titled **AI-Based Urban Waterlogging Risk Prediction and Preventive Response System**.

Our problem statement ID is **SIH26206**, and the theme of our project is **Disaster Management**.

The main idea of our project is to identify urban areas that have a higher risk of waterlogging and support authorities in taking preventive action.

Our solution combines rainfall information, historical waterlogging incidents and geospatial data to generate risk information for urban areas.

THANG LISH MEANING

Namma project oda main aim, urban area-la waterlogging risk adhigama irukkura places-ah identify panni, authorities advance-ah preventive action edukka support panradhu. Rainfall, historical incidents and geospatial data combine pannuvom.

PERSON 1 — SLIDE 2: IDEA TITLE

SPEAK

Urban waterlogging can affect roads, vulnerable locations, people and important facilities during heavy rainfall.

One major challenge is deciding which areas should receive preventive attention first, especially when manpower and budget are limited.

Our proposed system predicts waterlogging risk for **24, 48 and 72-hour horizons**.

It combines **rainfall data, historical incidents and geospatial data**.

The system then shows the predicted risk along with population and facility exposure, and recommends priority preventive actions.

It addresses the problem by identifying high-risk areas early, showing potentially affected people and facilities, prioritizing drains, roads and vulnerable areas, and supporting decisions with limited resources.

INNOVATION

The uniqueness of our solution is the workflow: **Predict → Explain → Prioritize → Optimize → Simulate**.

Predict means estimating waterlogging risk. Explain means showing the evidence behind the risk. Prioritize means considering impact. Optimize means planning based on available crews and budget. Simulate means testing different what-if scenarios.

IMPORTANT

Do not say the system guarantees flooding or gives 100% accurate results. Say **risk prediction** or **estimated risk**.

HANDOVER

Now I will hand over to my teammate to explain the technical approach and how our system works.

PERSON 2 — SLIDE 3: TECHNICAL APPROACH

SPEAK

Now I will explain how our system can be technically implemented.

Our frontend uses **Next.js, React, TypeScript and Tailwind CSS**. For the backend, we use **FastAPI and Python**. For storing spatial and application data, we use **PostgreSQL with PostGIS**.

For geospatial processing, we use **Leaflet and GeoPandas**. For data processing, we use **Pandas and NumPy**. Machine Learning is used for risk prediction. SHAP is used for explainability, and Google OR-Tools is used for optimization. Git and GitHub are used for version control.

FLOW CHART

The system starts with **rainfall, historical incidents and GIS data**.

First, the data goes through data processing. Then we perform geospatial analysis. After that, the system performs risk prediction.

Next, we analyse impact and exposure. Based on these results, we calculate a priority score. Then we perform crew and budget optimization.

We can also perform what-if simulations. Finally, the results are presented through a municipal dashboard.

IF ASKED BEFORE RAINFALL

When rainfall forecast information is available, it can be combined with historical incidents and geographical factors to estimate risk for different areas. The output is a risk estimate, not a guarantee.

HANDOVER

Next, I will explain the feasibility, challenges and mitigation strategies.

PERSON 2 — SLIDE 4: FEASIBILITY AND VIABILITY

FEASIBILITY

Our idea is technically feasible because it can use rainfall, historical incident and geospatial datasets.

The web-based architecture allows modular development. GIS, Machine Learning and optimization can be integrated into the same workflow. The system can also be scaled from a demonstration ward to larger urban areas.

CHALLENGES & RISKS

Local waterlogging data may be limited or inconsistent. Drainage and infrastructure data may be incomplete. Prediction uncertainty can occur because of data quality. Different cities may require different datasets and risk factors.

MITIGATION

We plan to validate and clean datasets before model use. We combine historical, spatial and rainfall features. We display supporting evidence with risk results. Models can be retrained and calibrated when new data becomes available. We also clearly distinguish real, public and synthetic or demonstration data.

HANDOVER

Now my teammate will explain the target users, impact and benefits of the solution.

PERSON 3 — SLIDE 5: IMPACT AND BENEFITS

TARGET USERS

The primary target users are **municipal authorities, disaster management teams, field response crews, ward engineers and citizens.**

IMPACT

The first impact is early identification of potential waterlogging risk. Second, it helps with better prioritization of vulnerable areas. Third, it provides visibility of population and critical-facility exposure. Finally, it can support more efficient use of limited crews and budgets.

EXAMPLE

For example, if a particular area is identified as a higher-risk area, authorities can use the available information to prioritize that location for preventive inspection or intervention.

BENEFITS

The solution supports **proactive rather than reactive action**. It can help reduce avoidable disruption during heavy rainfall, enable data-driven municipal decisions, and provide a scalable framework for urban waterlogging management.

IMPORTANT

Do not say the system guarantees that a specific area will flood. Say it identifies or estimates higher risk and helps prioritize preventive action.

PERSON 3 — SLIDE 6: REFERENCES + CONCLUSION

REFERENCES

For our research and data references, we considered several relevant sources.

The first is the **India Meteorological Department (IMD)**, relevant for rainfall and weather data.

The second is the **India Open Government Data Platform**, which provides a rainfall data catalogue.

The third is the **National Remote Sensing Centre / NDEM**, relevant for flood hazard and hydrological disaster data.

We also use **OpenStreetMap** as a geospatial reference for roads, waterways, amenities and other map information.

CONCLUSION

To conclude, our project aims to move urban waterlogging management from reactive response towards proactive prevention.

By combining rainfall, historical incident and geospatial information, the system can estimate risk, identify potentially vulnerable areas, prioritize preventive actions and support better resource allocation.

Our goal is to provide a practical decision-support system for urban authorities.

Thank you. We are happy to answer your questions.

QUICK Q&A;

What is the main problem?

Urban waterlogging risk needs to be identified early so authorities can prioritize preventive action.

What data do you use?

Rainfall, historical incidents and geospatial data. The PPT also includes population/facility exposure and infrastructure-related analysis.

How do you predict before rainfall?

Use available rainfall forecast information together with historical and geospatial features to estimate risk. It is not a guarantee.

Why 24/48/72 hours?

These are the prediction horizons stated in the proposed solution. Do not claim an accuracy unless it has been measured.

What if data is incomplete?

Validate and clean data, combine feature types, show supporting evidence, and retrain/calibrate models as new data becomes available.

Who uses it?

Municipal authorities, disaster management teams, field response crews, ward engineers and citizens.

What is unique?

Predict → Explain → Prioritize → Optimize → Simulate.